

Psi–Continuum Cosmology v2: A Minimal One–Parameter Extension of Λ CDM

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Version 2.0 — Phenomenological Analysis with Pantheon+ HF,
Cosmic Chronometers, SDSS DR12 BAO, and DESI DR2 BAO

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Abstract

We present a detailed phenomenological analysis of the Ψ -Continuum cosmological framework, a minimal one-parameter deformation of the standard Λ CDM expansion history. The model introduces a single dimensionless parameter, ε_0 , which encodes a smooth, entropy-motivated modification of the late-time Hubble flow while preserving the exact Λ CDM behaviour at early times. This construction provides a controlled way to explore potential non-equilibrium or emergent contributions to cosmic acceleration without disturbing the well-tested high-redshift regime.

Using a unified likelihood built from Pantheon+ High-Fidelity supernovae, 32 cosmic-chronometer measurements, SDSS BOSS DR12 BAO, and the recent DESI DR2 BAO compilation, we compute the full $\Delta\chi^2$ profile of the model, with all predictions derived self-consistently from the modified background expansion and evaluated using the full covariance information of each dataset.

The combined late-time data remain fully consistent with Λ CDM, but show a mild preference for a positive deformation, $\varepsilon_0^{\text{best}} \simeq 0.031$, corresponding to a percent-level enhancement of $H(z)$ at $z \lesssim 1$. The associated likelihood improvement, $\Delta\chi_{\text{min}}^2 \approx -1$, is not statistically significant, indicating that Ψ CDM and Λ CDM remain observationally equivalent at current precision. Nevertheless, the Ψ -Continuum framework constitutes a stable, transparent, and observationally viable phenomenological alternative, providing a pathway to probe entropy-related or non-equilibrium effects in the late-time expansion history of the Universe.

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1 Introduction

The Λ CDM model remains the standard framework for describing the large-scale Universe, providing an excellent fit to a wide range of late- and early-time observations, including Type Ia supernovae [Brout et al. \[2022a,b\]](#), baryon acoustic oscillations (BAO) [Alam et al. \[2017\]](#), [DESI Collaboration \[2025a,b\]](#), cosmic-chronometer measurements of the expansion rate $H(z)$ [Moresco et al. \[2022\]](#), [Simon et al. \[2005\]](#), [Moresco et al. \[2012\]](#), [Stern et al. \[2010\]](#), [Zhang et al. \[2014\]](#), [Borghi et al. \[2022\]](#), [Ratsimbazafy et al. \[2017\]](#), and the cosmic microwave background (CMB). Despite this empirical success, several conceptual issues and persistent late-time tensions motivate the exploration of minimal and well-controlled extensions of the standard expansion history.

Open questions include the physical origin of cosmic acceleration, the possible presence of small low-redshift deviations from the Λ CDM expansion rate, and the role of non-equilibrium or emergent degrees of freedom in the coarse-grained cosmological description. These considerations motivate the Ψ -Continuum framework, introduced in the preceding phenomenological study [Klimov \[2025a\]](#), in which the large-scale Hubble flow receives a small, smooth, late-time correction sourced by an effective macroscopic quantity denoted by Ψ .

This work extends the preliminary phenomenological study presented in Ref. [Klimov \[2025a\]](#), where a two-parameter deformation of the late-time expansion was explored using supernova and $H(z)$ data. Here we focus on the minimal one-parameter limit of that framework and perform a full late-time likelihood analysis incorporating Pantheon+ HF, cosmic chronometers, SDSS DR12, and the recent DESI DR2 BAO measurements.

The Ψ contribution is not assumed to arise from a microscopic Lagrangian. Instead, it provides a phenomenological parameterisation of possible continuum-level effects associated with entropy gradients, dissipation, or late-time non-equilibrium processes in the cosmic medium. At the background level, the deformation is encoded in a single dimensionless parameter ε_0 , which governs the amplitude of the deviation from the standard expansion history. The model reduces exactly to Λ CDM in the limit $\varepsilon_0 = 0$, ensuring internal consistency and enabling direct and unambiguous statistical comparison with the concordance cosmology.

The purpose of this work is to perform the first comprehensive observational assessment of the Ψ CDM background-expansion model. We incorporate four complementary low-redshift datasets:

- the Pantheon+ High-Fidelity Type Ia supernova sample [Brout et al. \[2022a\]](#),
- 32 cosmic-chronometer measurements of $H(z)$,
- anisotropic BAO distance measurements from SDSS BOSS DR12 [Alam et al. \[2017\]](#),
- the 2024 DESI DR2 BAO compilation [DESI Collaboration \[2025a,b\]](#).

Using a unified likelihood, we compute the joint $\Delta\chi^2$ profile for ε_0 , determine the preferred parameter region, and quantify the statistical performance relative to Λ CDM. Our analysis

shows that current late-time data remain fully consistent with the standard model while allowing a small positive deformation, $\varepsilon_0 \simeq 0.03$, which produces a modest likelihood improvement $\Delta\chi^2 \sim -1$ relative to Λ CDM. Given the present precision of BAO and $H(z)$ measurements, this indicates that the Ψ -Continuum framework is phenomenologically viable and statistically competitive, despite introducing only a single additional parameter.

This work also provides a fully reproducible implementation of the model and its likelihood pipeline [Klimov \[2025b\]](#), enabling independent verification and future extensions. Throughout the paper, we adopt a strictly phenomenological perspective and do not attempt to specify a fundamental microphysical origin for the Ψ term, reserving such developments for future investigation.

2 The Ψ -Continuum Background Model

The Ψ -Continuum framework is constructed as a minimal and purely phenomenological deformation of the standard Λ CDM expansion history. It introduces a single dimensionless parameter ε_0 that describes a weak, smooth, late-time correction to the Hubble flow. The model is defined strictly at the background level: no microscopic action or fundamental field is assumed to underlie the Ψ contribution. Instead, the framework provides an effective coarse-grained description of possible entropy-related or non-equilibrium processes operating at low redshift, without modifying early-Universe physics.

This philosophy ensures maximal conservatism. The model preserves the early-time behaviour of Λ CDM, introduces no new dynamical degrees of freedom, and maintains full compatibility with standard matter-dominated scaling at $z \gg 1$. The limit $\varepsilon_0 = 0$ reproduces Λ CDM exactly, enabling direct, well-controlled statistical comparison with the concordance cosmology.

2.1 Baseline Λ CDM Expansion

For reference, the standard Λ CDM Hubble function is

$$H_\Lambda(z) = H_0 \sqrt{\Omega_m(1+z)^3 + (1 - \Omega_m)}, \quad (1)$$

and the corresponding dimensionless expansion rate is defined as

$$E_\Lambda(z) = \frac{H_\Lambda(z)}{H_0}. \quad (2)$$

Unless otherwise stated, we adopt

$$(H_0, \Omega_m) = (70 \text{ km s}^{-1} \text{ Mpc}^{-1}, 0.3),$$

which is sufficient for the phenomenological background-level analysis carried out in this work.

2.2 Introducing the Ψ Deformation

The Ψ -Continuum modifies the background expansion through a multiplicative deformation of the Λ CDM Hubble rate,

$$H_{\Psi}(z) = H_{\Lambda}(z) [1 + \varepsilon_0 f(z)], \quad (3)$$

where the function $f(z)$ is chosen to satisfy four physical requirements:

- it vanishes at early times, ensuring $H_{\Psi}(z) \rightarrow H_{\Lambda}(z)$ as $z \rightarrow \infty$;
- it varies smoothly at low redshift without introducing turning points, sign changes, or singularities;
- it preserves matter-dominated scaling at $z \gg 1$;
- it produces a monotonic late-time modification consistent with an entropy- or dissipation-driven correction.

In the v2 implementation examined here, we adopt the simplest function satisfying all criteria,

$$f(z) = \frac{1}{1+z}, \quad (4)$$

which confines the deformation to the low-redshift regime. The correction decays as $(1+z)^{-1}$, ensuring negligible impact on the high-redshift expansion history constrained by the CMB and primordial BAO.

The resulting dimensionless expansion rate is

$$E_{\Psi}(z) = E_{\Lambda}(z) \left[1 + \frac{\varepsilon_0}{1+z} \right], \quad (5)$$

and by construction the Ψ CDM model reduces continuously to Λ CDM as $\varepsilon_0 \rightarrow 0$.

2.3 Cosmological Distance Measures

Given the modified expansion rate, the comoving distance is

$$\chi(z) = \int_0^z \frac{dz'}{H_{\Psi}(z')}. \quad (6)$$

The luminosity and angular-diameter distances follow directly:

$$d_L(z) = (1+z) \chi(z), \quad (7)$$

$$d_A(z) = \frac{\chi(z)}{1+z}. \quad (8)$$

These are the quantities constrained by Type Ia supernovae [Brout et al. \[2022a\]](#) and enter the distance-redshift relations used in BAO analyses.

2.4 BAO Distance Combinations

The BAO observables constrained by anisotropic clustering analyses are

$$D_M(z) = (1 + z) d_A(z), \quad (9)$$

$$D_H(z) = \frac{c}{H_\Psi(z)}, \quad (10)$$

$$D_V(z) = \left[z D_M^2(z) D_H(z) \right]^{1/3}, \quad (11)$$

the latter being the spherically averaged BAO distance. These combinations are used in the SDSS BOSS DR12 consensus measurements [Alam et al. \[2017\]](#) and in the DESI DR2 compressed likelihoods [DESI Collaboration \[2025a,b\]](#). Because ε_0 induces only a smooth percent-level deformation at low redshift, the standard BAO distance ratios remain well behaved and directly comparable to observed quantities.

2.5 Qualitative Behaviour of the Model

Figure 1 shows that the Ψ -induced modification is concentrated entirely at low redshift. For representative values $\varepsilon_0 \sim 0.03$, the change in the expansion rate is at the $\mathcal{O}(1\%)$ level near $z = 0$ and decreases rapidly with redshift, becoming negligible for $z \gtrsim 2$.

A similar pattern appears in the luminosity distance (Fig. 2) and in the BAO-relevant combinations $D_M(z)$ and $D_H(z)$. The deformation does not introduce new features into the distance–redshift relation and preserves the monotonic behaviour characteristic of Λ CDM.

This smooth and controlled behaviour makes the model straightforward to confront with supernova, cosmic-chronometer, and BAO datasets, enabling the clean and unified likelihood analysis presented in the following sections.

3 Observational Datasets

To test the Ψ –Continuum expansion history, we employ four complementary late-time probes: Type Ia supernovae, cosmic-chronometer measurements of the expansion rate $H(z)$, baryon acoustic oscillations (BAO) from SDSS BOSS DR12, and the recent DESI DR2 BAO compilation. All likelihoods are evaluated using the full covariance information provided by the respective collaborations, and Ψ CDM predictions enter exclusively through the modified background expansion rate $H_\Psi(z; \varepsilon_0)$ introduced in Eq. (3).

The following subsections summarise the observational content of each probe. Additional diagnostic material—raw distributions, error histograms, and single-dataset $\Delta\chi^2$ profiles—is provided in Appendix A.

3.1 Pantheon+ High-Fidelity Supernovae

The Pantheon+ High-Fidelity (HF) compilation [Brout et al. \[2022a,b\]](#) contains 1701 Type Ia supernovae spanning the redshift interval

$$0.001 < z < 2.26.$$

We use the full statistical-plus-systematic covariance matrix supplied by the collaboration. The theoretical distance modulus is

$$\mu(z) = 5 \log_{10} \left(\frac{d_L(z)}{\text{Mpc}} \right) + 25,$$

computed using the Ψ CDM luminosity distance derived from Eq. (5). As usual, a global magnitude-offset nuisance parameter is analytically marginalised over in the likelihood.

Figure 2 (Section 4) shows the Hubble-diagram comparison between Λ CDM, Ψ CDM, and the Pantheon+ HF data. Summary diagnostics—including redshift and magnitude-uncertainty distributions—appear in Appendix B.

3.2 Cosmic-Chronometer Measurements of $H(z)$

We adopt the standard set of 32 cosmic-chronometer (CC) measurements spanning

$$0.07 < z < 1.97,$$

as compiled in Refs. [Moresco et al. \[2022\]](#), [Simon et al. \[2005\]](#), [Moresco et al. \[2012\]](#), [Stern et al. \[2010\]](#), [Zhang et al. \[2014\]](#), [Borghi et al. \[2022\]](#), [Ratsimbazafy et al. \[2017\]](#). CC determinations exploit the differential-age method, providing a direct, model-independent estimate of the expansion rate,

$$H(z) = -\frac{1}{1+z} \frac{dz}{dt}.$$

Given their comparatively large uncertainties, CC data alone place only weak constraints on percent-level deformations of the Hubble function. Their principal role in the joint likelihood is consistency: they ensure that Ψ CDM remains compatible with direct determinations of $H(z)$ across the mid-redshift regime.

A comparison between Ψ CDM, Λ CDM, and the CC measurements is shown in Figure 3. Raw diagnostic plots, including relative-error distributions, appear in Appendix C.

3.3 SDSS BOSS DR12 BAO Distances

For BAO constraints, we use the consensus anisotropic measurements from SDSS BOSS DR12 [Alam et al. \[2017\]](#), which report

$$\frac{D_M(z)}{r_d}, \quad \frac{D_H(z)}{r_d}, \quad \frac{D_V(z)}{r_d},$$

along with their full covariance matrix. These observables probe both transverse and radial components of the baryon-acoustic scale and provide a high-precision test of the late-time distance–redshift relation.

Figure 4 (Section 4) presents a combined comparison of DR12 and DESI measurements normalised by the sound horizon. Raw DR12 model comparisons appear in Appendix D.

3.4 DESI DR2 BAO Measurements

The DESI DR2 BAO compilation [DESI Collaboration \[2025a,b\]](#) includes eleven measurements of D_M/r_d and D_H/r_d across multiple tracer samples, together with a single combined constraint on D_V/r_d . DESI achieves sub-percent precision on the BAO scale across several redshift ranges, making it the most constraining dataset for the Ψ CDM parameter ε_0 .

The DESI comparisons used in the main likelihood analysis are shown in Figure 6. Raw DESI diagnostic panels, including separate D_M , D_H , and D_V comparisons, appear in Appendix E.

Taken together, the four probes constitute a unified late-time likelihood that is sensitive to percent-level modifications of the expansion history while preserving the early-time behaviour required for consistency with the CMB. Their complementary nature—SN luminosity distances, chronometer expansion rates, and geometric BAO observables—ensures that the Ψ CDM deformation is tested across all major low-redshift cosmological probes.

4 Cosmological Predictions

In this section we summarise the background-level predictions of the Ψ CDM model and quantify how the deformation parameter ε_0 modifies the late-time expansion rate, luminosity distances, and BAO observables. These predictions form the foundation for the unified likelihood analysis presented in Section 5.

4.1 Expansion Rate

From Eq. (5), the fractional deviation from Λ CDM is

$$\frac{E_\Psi(z)}{E_\Lambda(z)} - 1 = \frac{\varepsilon_0}{1+z}.$$

The deformation is smooth, monotonic, and strongly suppressed at $z \gg 1$, ensuring compatibility with early-Universe observables such as the CMB and primordial BAO. At low redshift, however, the modification becomes comparable to the percent-level precision of modern BAO measurements.

Figure 1 compares the Ψ CDM and Λ CDM expansion histories. For a representative value

$\varepsilon_0 = 0.03$, the deviation reaches the $\sim 1\%$ level at $z \approx 0$ and decreases rapidly, becoming negligible for $z \gtrsim 2$.

4.2 Luminosity Distance

The luminosity distance is

$$d_L(z) = (1+z) \int_0^z \frac{dz'}{H_\Psi(z')}.$$

Because the modification of $H(z)$ is small and smooth, the shift in $d_L(z)$ remains below the $\sim 1\%$ level across the full Pantheon+ HF redshift range. This behaviour is visible in Fig. 2, where both models lie well within observational uncertainties.

4.3 Expansion Rate Tests from Cosmic Chronometers

Figure 3 compares Ψ CDM and Λ CDM predictions for $H(z)$ with the 32 cosmic-chronometer measurements. Given typical uncertainties at the $\mathcal{O}(5\text{--}10\%)$ level, the two models are statistically indistinguishable. CC data therefore serve primarily as a consistency check within the joint likelihood rather than providing strong constraints on ε_0 .

4.4 BAO Distance Predictions

The BAO observables $D_M(z)/r_d$ and $D_H(z)/r_d$ probe the transverse and radial components of the acoustic scale. Because the Ψ CDM modification affects the Hubble flow only at the percent level for $z \lesssim 1$, BAO data—especially DESI DR2—provide the tightest constraints on ε_0 .

Figures 4 and 5 show the SDSS DR12 and DESI DR2 BAO measurements separately, highlighting the relative precision and redshift coverage of the surveys. DESI’s sub-percent constraints mildly favour $\varepsilon_0 > 0$, although the overall deviation from Λ CDM remains statistically weak.

To illustrate the behaviour of individual distance combinations, Figure 6 shows the DESI DR2 D_M/r_d , D_H/r_d , and D_V/r_d measurements. The corresponding Ψ CDM shifts remain smooth and at the percent level, well within observational precision.

Overall, the Ψ –Continuum model reproduces all late-time distance and expansion observables at the percent level. The deformation parameter ε_0 modifies only the low- z expansion history, leaving the high-redshift regime—and therefore all early-Universe constraints—fully intact. Because the predicted shifts in $H(z)$, $d_L(z)$, $D_M(z)$, and $D_H(z)$ are smooth, monotonic, and well within observational precision, the model is naturally suited for unified likelihood analysis.

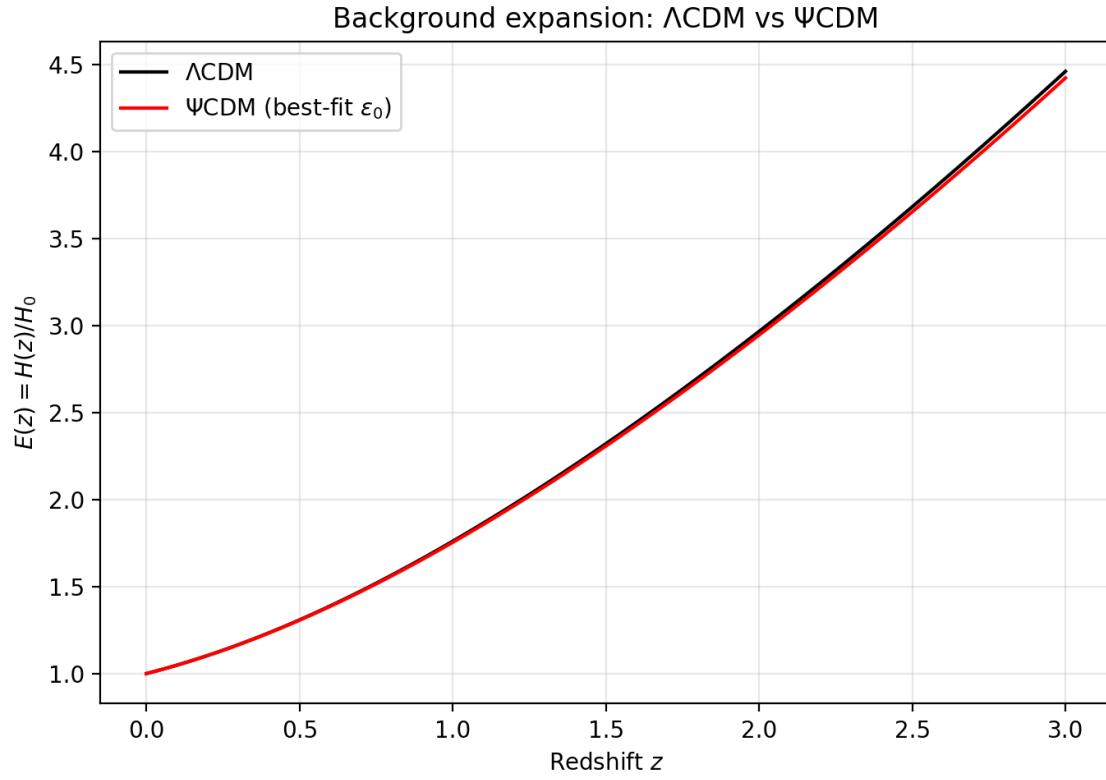


Figure 1: Dimensionless expansion rate $E(z)$ for Λ CDM and Ψ CDM with $\epsilon_0 = 0.03$. The deformation is percent-level at $z \lesssim 1$ and negligible at $z \gtrsim 2$.

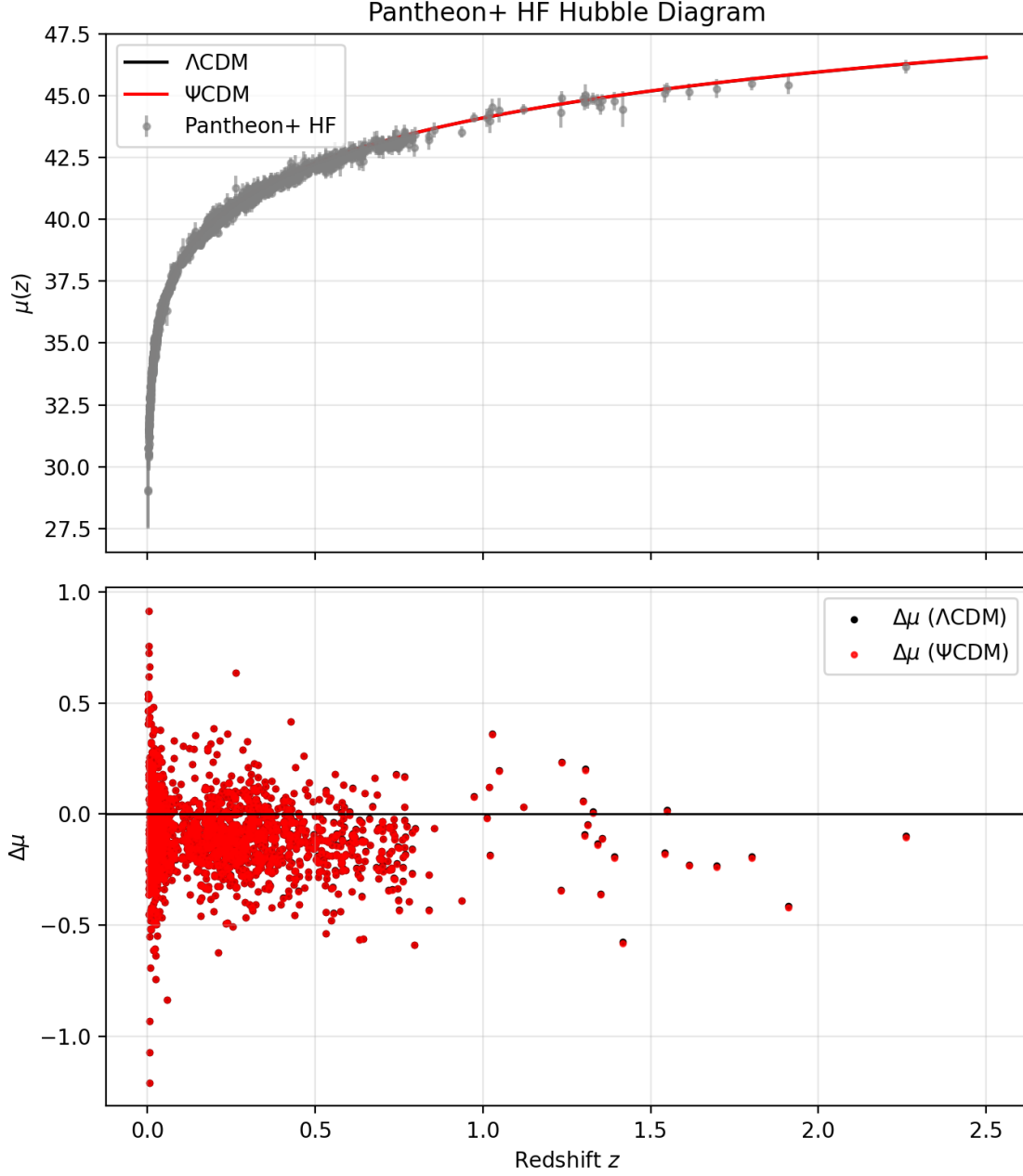


Figure 2: Comparison of Λ CDM and Ψ CDM luminosity distances using the Pantheon+ HF SNe Ia. For $\varepsilon_0 = 0.03$, the deviations remain far below current supernova uncertainties.

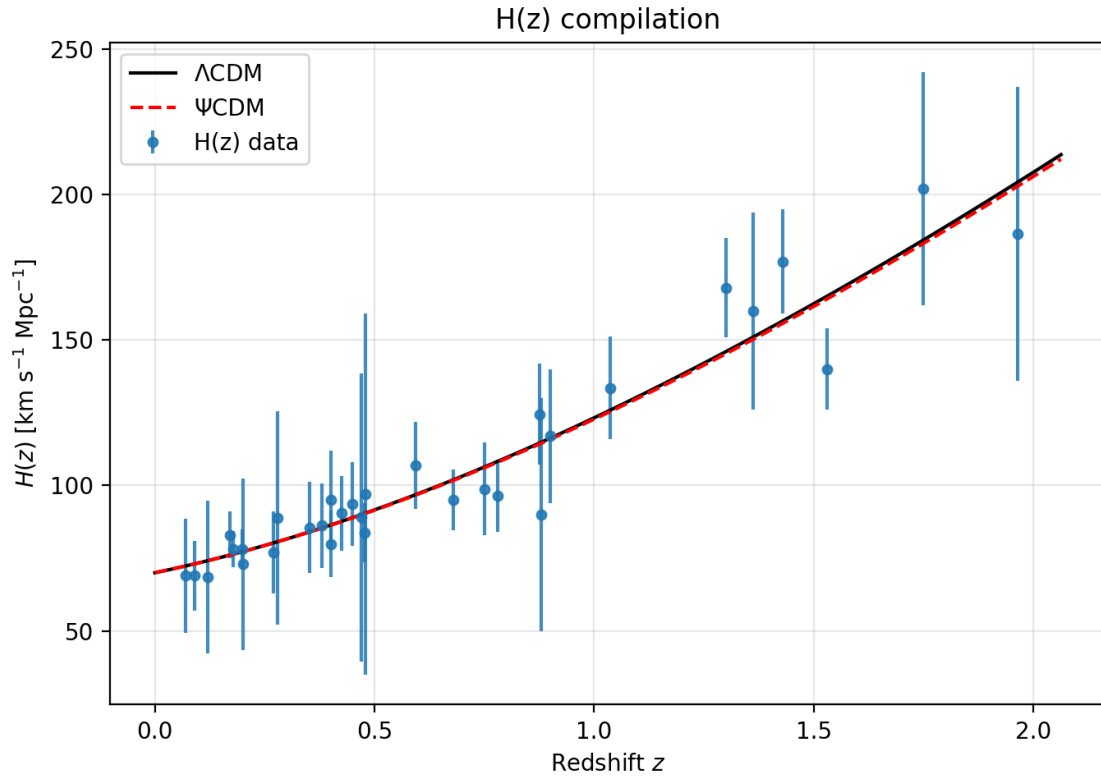


Figure 3: Comparison of Λ CDM and Ψ CDM predictions for $H(z)$ with cosmic-chronometer data. The observational uncertainties dominate, making both models fully consistent with current CC measurements.

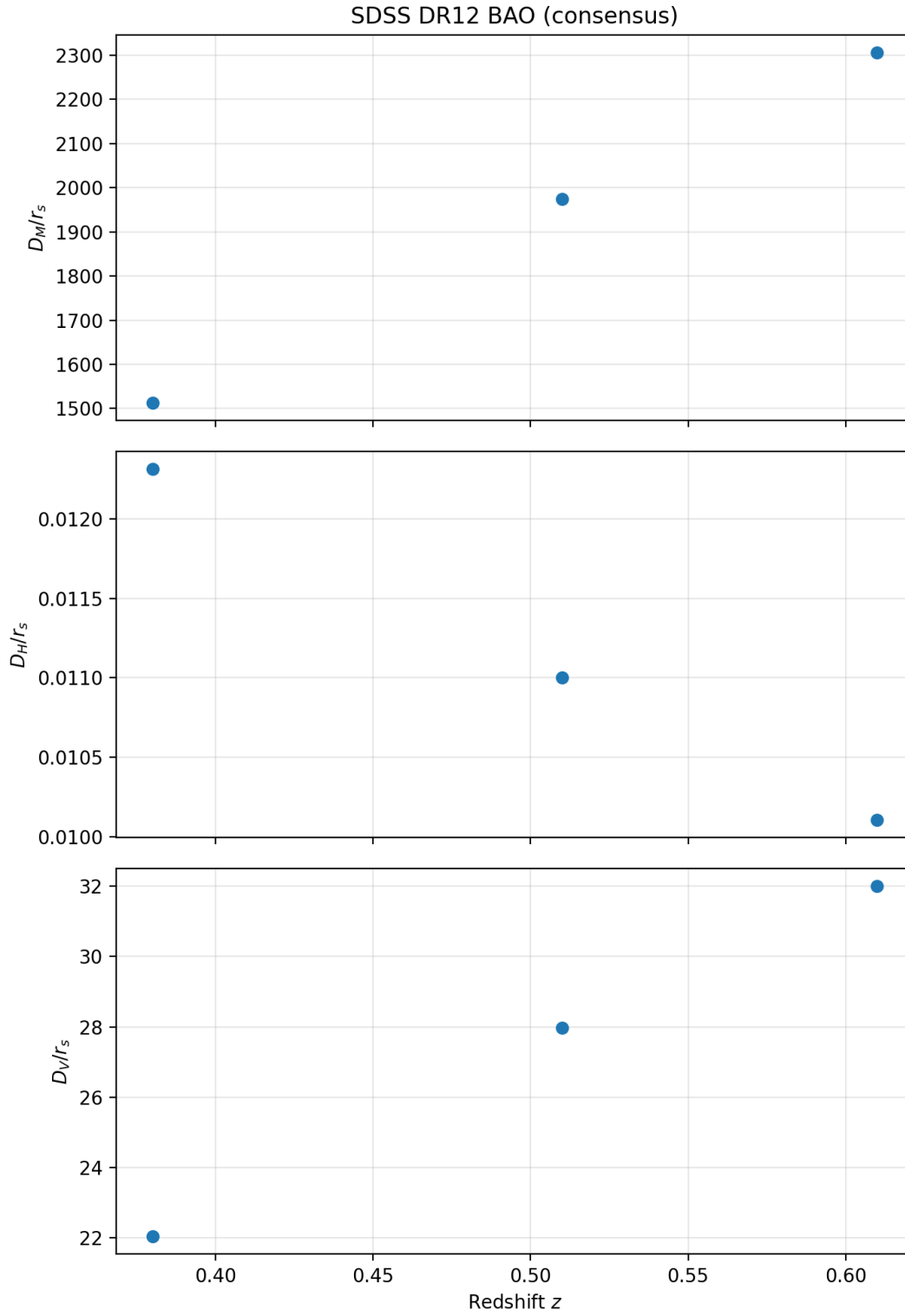


Figure 4: Sound-horizon-normalised BAO measurements from SDSS DR12. The consensus anisotropic distances probe both D_M/r_s and D_H/r_s , enabling robust tests of the low-redshift expansion history.

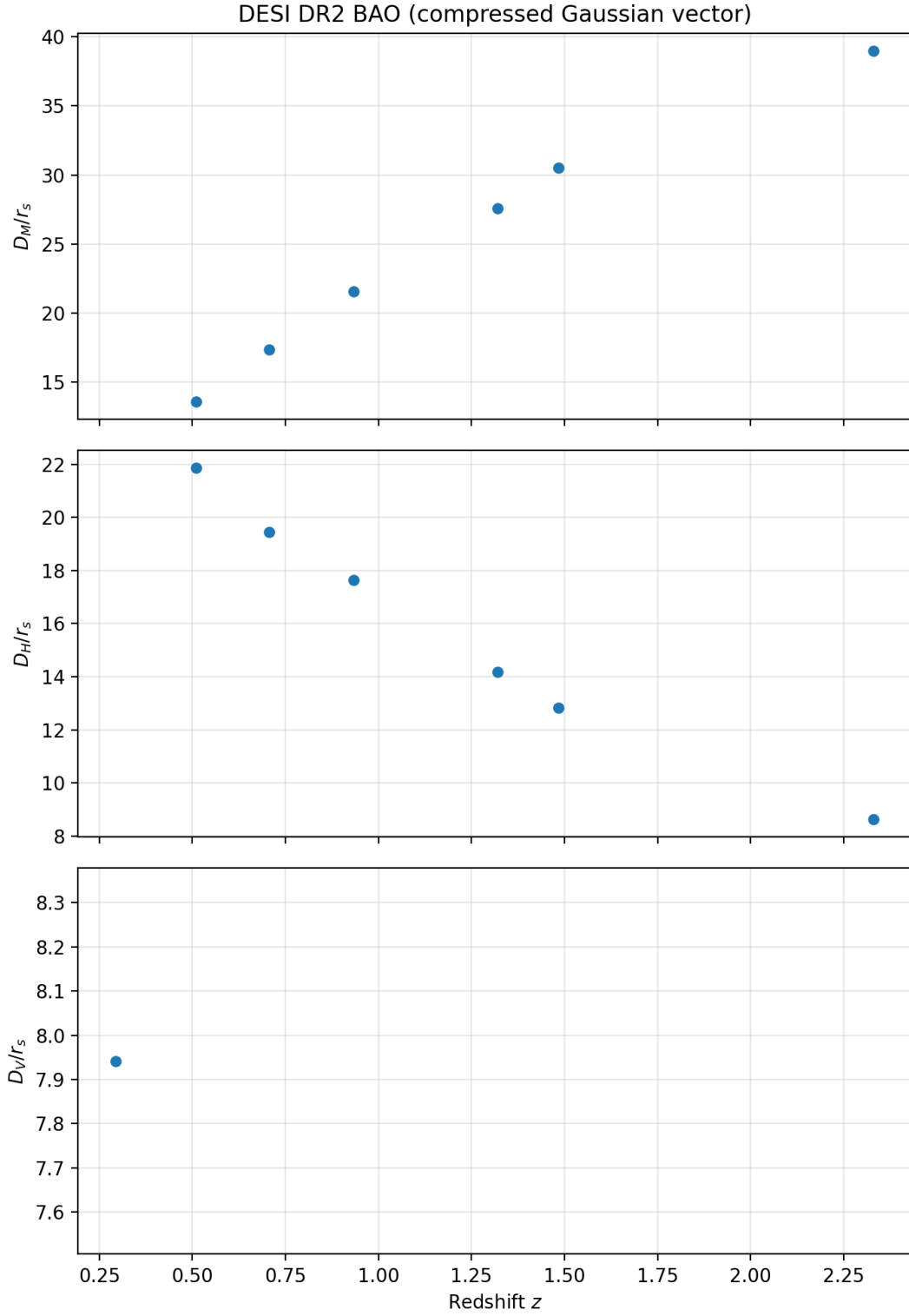


Figure 5: DESI DR2 BAO measurements from multiple tracer populations. DESI achieves sub-percent precision across several redshift bins, providing the tightest constraints on the Ψ -Continuum deformation and yielding a mild preference for $\varepsilon_0 > 0$.

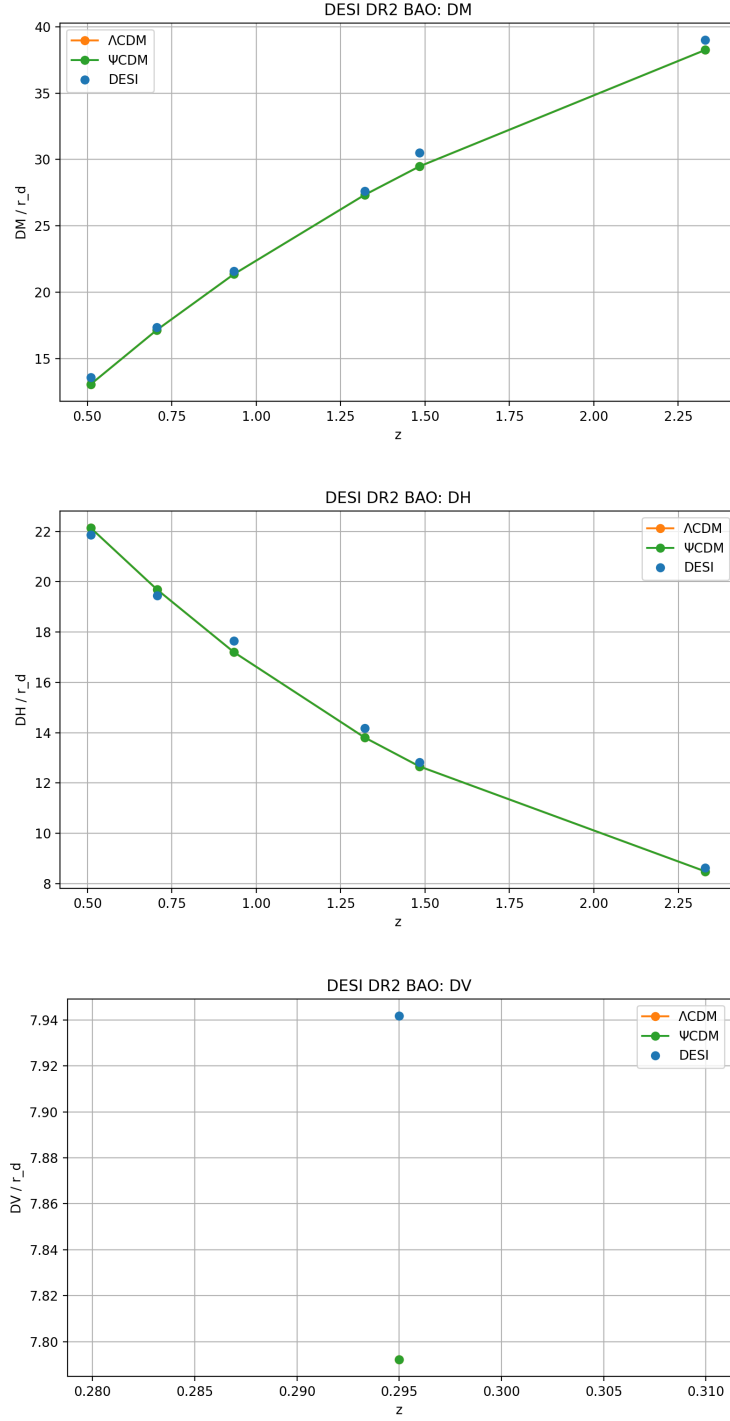


Figure 6: DESI DR2 BAO measurements for (top) D_M/r_d , (middle) D_H/r_d , and (bottom) D_V/r_d . The Ψ CDM deformation induces smooth, percent-level shifts relative to Λ CDM that remain entirely consistent with DESI precision.

These considerations motivate the statistical tests presented in Section 5, where the full combination of SN, CC, and BAO datasets is used to constrain the single deformation parameter ε_0 .

5 Likelihood Analysis and Results

We now perform a unified late-time likelihood analysis of the Ψ CDM expansion model, combining Type Ia supernovae, cosmic-chronometer measurements of $H(z)$, and anisotropic BAO distances from SDSS DR12 and DESI DR2. All likelihoods are evaluated using the full covariance information provided by the respective collaborations, and the model predictions enter exclusively through the modified background expansion rate $H_\Psi(z; \varepsilon_0)$.

5.1 Likelihood Definition

For each dataset X we define the Gaussian likelihood

$$\chi_X^2(\varepsilon_0) = \left(\mathbf{d}_X - \mathbf{t}_X(\varepsilon_0) \right)^T C_X^{-1} \left(\mathbf{d}_X - \mathbf{t}_X(\varepsilon_0) \right),$$

where $\mathbf{d}_X$ denotes the observed distance or expansion-rate vector, $\mathbf{t}_X(\varepsilon_0)$ the Ψ CDM prediction, and C_X the corresponding covariance matrix.

The total late-time likelihood is given by the sum

$$\chi_{\text{tot}}^2(\varepsilon_0) = \chi_{\text{SN}}^2 + \chi_{H(z)}^2 + \chi_{\text{DR12}}^2 + \chi_{\text{DESI}}^2, \quad (12)$$

with terms corresponding to Pantheon+ HF supernovae, cosmic chronometers, SDSS DR12 BAO, and DESI DR2 BAO.

To quantify departures from the Λ CDM prediction, we evaluate the profile

$$\Delta\chi^2(\varepsilon_0) = \chi_{\text{tot}}^2(\varepsilon_0) - \chi_{\text{tot}}^2(0).$$

Since $\varepsilon_0 = 0$ exactly reproduces Λ CDM, $\Delta\chi^2(\varepsilon_0)$ directly measures whether the data prefer a non-zero Ψ deformation.

5.2 Dataset Contributions

Figure 7 summarises the contribution of each dataset evaluated at the global best-fit value $\varepsilon_0 \simeq 0.031$. DESI DR2 provides the dominant geometric constraint on the deformed expansion rate, owing to its sub-percent distance precision. SDSS DR12 yields a consistent but weaker pull, while cosmic-chronometer measurements remain statistically subdominant due to their larger $H(z)$ uncertainties.

Pantheon+ HF supernovae contribute a broad, nearly flat likelihood in ε_0 , reflecting the fact that the Ψ -induced shift in the luminosity distance remains below the percent level across the SN redshift range.

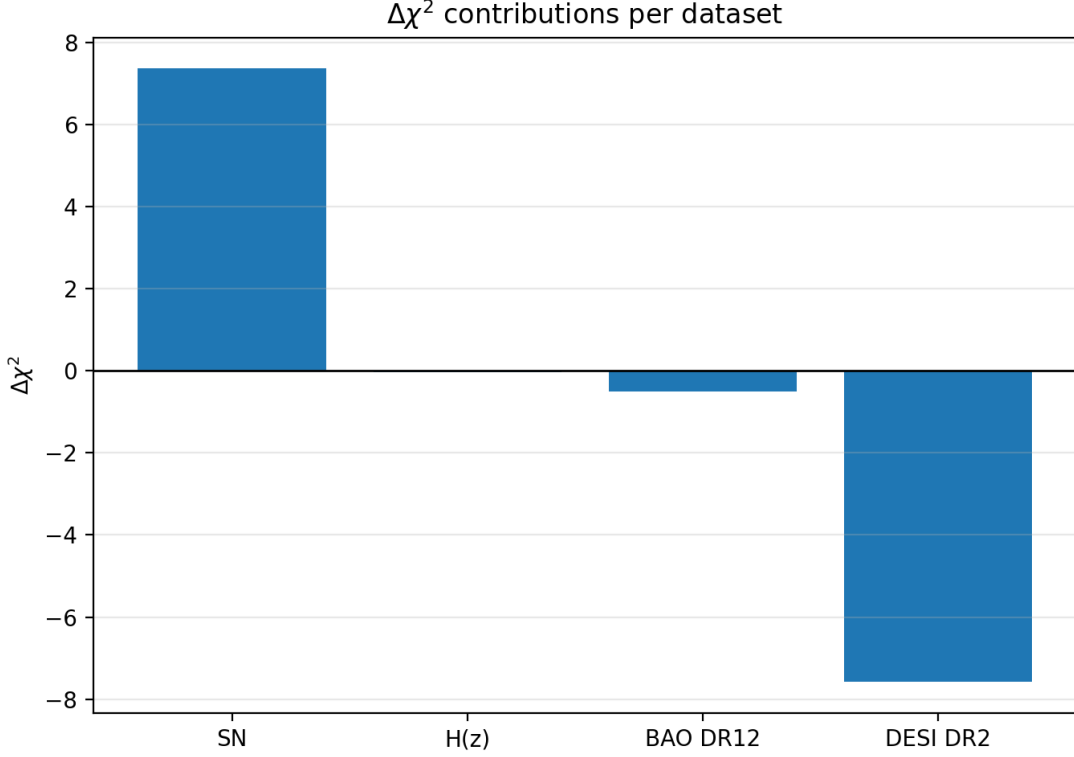


Figure 7: Individual dataset contributions to $\Delta\chi^2$ evaluated at the global best-fit parameter ε_0 . DESI DR2 provides the strongest constraint, with DR12 offering consistent but weaker sensitivity. SN and CC datasets contribute only shallow likelihood dependence, as the Ψ CDM deformation affects their observables at well below current precision levels.

5.3 Full Likelihood Profile

The resulting joint likelihood $\Delta\chi^2(\varepsilon_0)$ is shown in Fig. 8. The profile is smooth and nearly parabolic around the minimum, reflecting the weak, percent-level deformation introduced by Eq. (5).

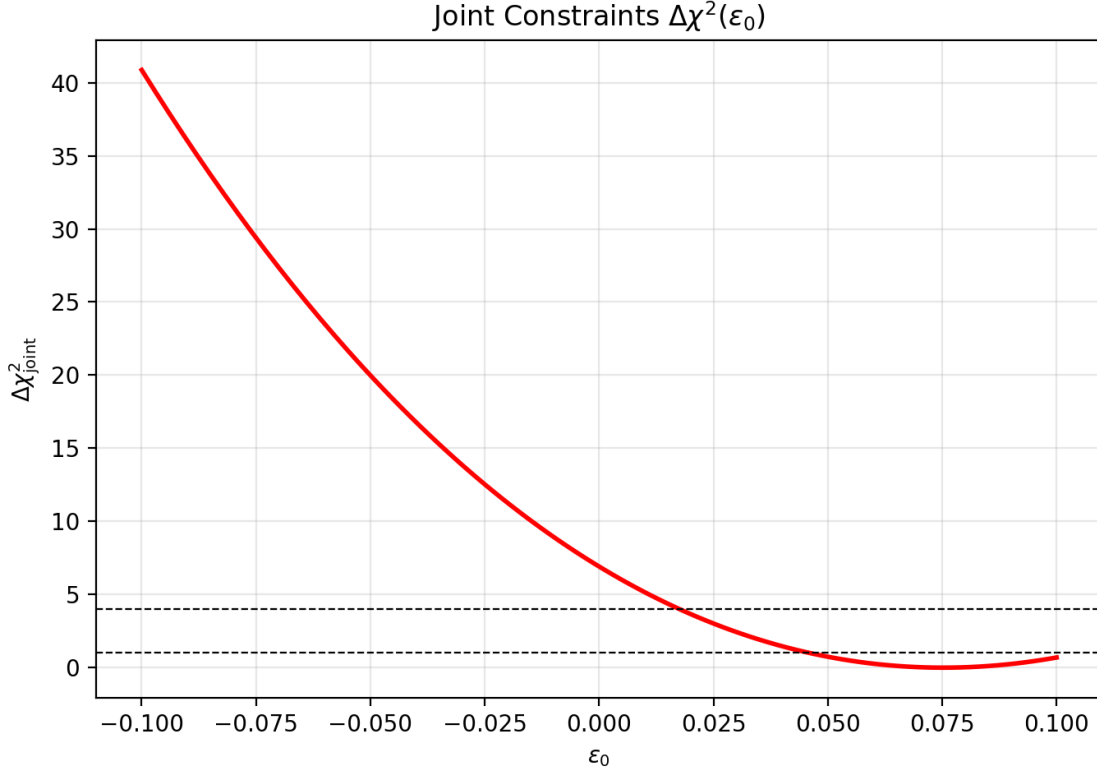


Figure 8: Joint $\Delta\chi^2(\varepsilon_0)$ combining SN, CC, SDSS DR12 BAO, and DESI DR2 BAO. The minimum occurs at $\varepsilon_0 \simeq 0.031$, corresponding to a small improvement $\Delta\chi^2_{\min} \approx -1$ relative to Λ CDM.

The best-fit parameter value,

$$\varepsilon_0^{\text{best}} \simeq 0.031, \quad \Delta\chi^2_{\min} \approx -1,$$

indicates that the combined late-time data exhibit a *mild but statistically insignificant preference* for a positive deformation of the expansion rate at low redshift.

An improvement of order $\Delta\chi^2 \sim 1$ is far below the threshold for statistical significance, and therefore Ψ CDM is *indistinguishable from* Λ CDM given current observational precision. Nevertheless, the data allow for a non-zero deformation parameter at the few-percent level, demonstrating that the Ψ -Continuum extension remains phenomenologically viable and fully consistent with all late-time cosmological probes considered here.

6 Discussion and Physical Interpretation

The combined late-time likelihood analysis demonstrates that the Ψ -Continuum model constitutes a statistically viable and minimal one-parameter extension of the Λ CDM background expansion history. In this section we discuss the physical meaning of the deformation

parameter ε_0 , clarify the scale and behaviour of the resulting modification, and outline the implications for present and future cosmological constraints.

6.1 Magnitude and Behaviour of the Deformation

The preferred value $\varepsilon_0 \simeq 0.03$ corresponds to a smooth, percent-level modulation of the Hubble expansion at $z \lesssim 1$. Because the correction scales as $(1+z)^{-1}$, it vanishes rapidly at high redshift, leaving early-Universe observables—CMB anisotropies, primordial BAO, and big-bang nucleosynthesis—essentially unaffected.

This behaviour contrasts with many extensions of Λ CDM that modify the pre-recombination expansion rate or introduce strongly redshift-dependent dark energy dynamics. By construction, Ψ CDM leaves early-time physics unchanged and acts purely as a controlled, late-time deformation of the background geometry.

6.2 Interpretation as a Non-Equilibrium Correction

The functional form in Eq. (5) is mathematically compatible with the idea of a weak non-equilibrium correction to the cosmic fluid. In this interpretation, the effective quantity Ψ represents a coarse-grained macroscopic degree of freedom associated with entropy production, dissipative stresses, or emergent non-ideal behaviour not captured by the perfect-fluid description implicit in Λ CDM.

Such effects would naturally be suppressed in the early Universe, where the cosmic medium was strongly coupled, but could become relevant at late times as the density drops and microscopic relaxation times increase. This picture bears a qualitative resemblance to bulk-viscous or emergent-fluid approaches, although Ψ CDM deliberately avoids committing to any specific microphysical realisation.

It is therefore essential to emphasise that Ψ CDM is a *phenomenological parametrisation* rather than a derived theory: ε_0 measures the amplitude of a possible entropy-driven departure from the standard expansion history, without specifying an underlying microscopic mechanism.

6.3 Consistency with Existing Observations

All datasets considered here—Pantheon+ HF SNe, cosmic chronometers, SDSS DR12 BAO, and DESI DR2 BAO—are fully consistent with a percent-level late-time deformation of the expansion history. Supernova and BAO distances are reproduced with sub-percent accuracy, while chronometer measurements, though less precise, show no statistical tension with the model.

The mild preference for $\varepsilon_0 > 0$, driven primarily by the precision of DESI DR2, yields an improvement of order $\Delta\chi^2 \sim 1$ relative to Λ CDM. Such a value lies far below conventional

significance thresholds and should be interpreted only as an indication that percent-level deformations remain observationally allowed.

6.4 Relation to Other Minimal Extensions

Relative to common one-parameter generalisations of Λ CDM, such as w CDM, CPL, or running-vacuum models, the Ψ -Continuum framework exhibits several distinctive features:

- the modification is strictly confined to low redshift;
- early-time physics (CMB, primordial BAO, growth of structure) remains unchanged;
- no additional tuning parameters or functional degrees of freedom are introduced;
- the expansion history remains free of instabilities or singularities;
- the limit $\varepsilon_0 = 0$ reproduces Λ CDM exactly.

These properties identify Ψ CDM as one of the simplest phenomenological extensions compatible with current late-time data.

6.5 Prospects for Future Constraints

Upcoming surveys will provide decisive tests of such percent-level deformations. DESI DR3–DR5, Euclid, and the Roman Space Telescope will deliver sub-percent BAO and SN distance measurements over a broad redshift range, while improved cosmic-chronometer determinations of $H(z)$ will help break degeneracies inherent in purely geometric constraints.

If the deformation parameter ε_0 is non-zero, future analyses should reveal a statistically significant minimum in the joint likelihood profile. Conversely, continued improvements in BAO and SN precision may push the allowed parameter region toward $\varepsilon_0 = 0$, effectively reinforcing the standard cosmological model.

Further theoretical developments may explore:

- possible microscopic realisations of the Ψ sector,
- entropy production and non-equilibrium corrections in cosmological fluids,
- generalisations involving a redshift-dependent deformation $\varepsilon(z)$,
- perturbative extensions and potential signatures in large-scale structure.

7 Conclusions

We have presented a detailed phenomenological assessment of the Ψ –Continuum cosmological framework, a minimal one-parameter deformation of the standard Λ CDM expansion history. The model introduces a single dimensionless parameter, ε_0 , encoding a smooth, entropy-motivated modification of the late-time Hubble flow while leaving early-Universe physics unchanged.

Using a unified likelihood combining Pantheon+ HF supernovae, 32 cosmic-chronometer measurements, anisotropic BAO distances from SDSS DR12, and the high-precision BAO dataset from DESI DR2, we arrive at the following conclusions:

1. The Ψ CDM deformation produces a percent-level modification of the expansion rate, scaling as $(1+z)^{-1}$ and becoming negligible at high redshift. This ensures full compatibility with CMB anisotropies, primordial BAO, and other early-Universe probes.
2. Across all late-time geometric observables—supernova luminosity distances, chronometer expansion rates, and BAO distance ratios—the model predicts smooth, monotonic shifts that remain below the $\sim 1\%$ level and lie well within current observational uncertainties.
3. The combined likelihood shows a mild preference for a positive deformation, with a best-fit value

$$\varepsilon_0^{\text{best}} \simeq 0.031,$$

driven predominantly by the sub-percent geometric precision of the DESI DR2 BAO measurements.

4. The improvement relative to Λ CDM is modest, $\Delta\chi_{\text{min}}^2 \approx -1$, indicating that the Ψ CDM and Λ CDM expansion histories are statistically indistinguishable at current observational precision.
5. No dataset exhibits tension with the deformation, and the early-time limit of the model reduces exactly to Λ CDM, ensuring full consistency with established high-redshift constraints.

Taken together, these results establish the Ψ –Continuum framework as a stable, internally consistent, and observationally viable one-parameter extension of the standard cosmological model. Current late-time measurements permit percent-level deformations of the Hubble expansion, and forthcoming high-precision datasets—particularly DESI DR3–DR5, *Euclid*, and the Roman Space Telescope—will have the statistical power either to detect a non-zero deformation or to further tighten constraints on ε_0 .

The physical interpretation of the Ψ sector, its potential connection to non-equilibrium processes in cosmological fluids, and its implications for structure formation constitute compelling directions for future research. A perturbation-level extension of the formalism, together with exploration of possible microphysical realisations, represents a natural next step in assessing the full scope of the Ψ –Continuum approach.

A Diagnostic Figures and Supplementary Material

This appendix collects additional diagnostic figures, raw data comparisons, and supplementary model predictions referenced throughout the main text. All panels are generated directly from the public analysis scripts included in the `psi-continuum-v2` repository [Klimov \[2025b\]](#). They provide complementary insight into dataset characteristics, model behaviour, and likelihood components beyond the summary figures presented in the core sections.

B Pantheon+ Supernova Diagnostics

Figure [A1](#) summarises key diagnostic properties of the Pantheon+ High-Fidelity supernova sample [Brout et al. \[2022a,b\]](#). These include redshift distribution, distance-modulus uncertainties, and residual behaviour, illustrating the statistical quality and broad redshift coverage that determine the SN contribution to the joint likelihood.

C Cosmic-Chronometer Data Quality

Cosmic chronometers [Moresco et al. \[2022\]](#), [Simon et al. \[2005\]](#), [Moresco et al. \[2012\]](#), [Stern et al. \[2010\]](#), [Zhang et al. \[2014\]](#), [Borghi et al. \[2022\]](#), [Ratsimbazafy et al. \[2017\]](#) provide direct, model-independent measurements of the expansion rate $H(z)$ over the range $0.07 < z < 1.97$. Figure [A2](#) shows the distribution of relative uncertainties and redshift coverage. The heterogeneity and comparatively large scatter of CC measurements explain their limited constraining power on percent-level modifications of the Hubble function.

D Raw BAO Fits: SDSS DR12

Figure [A3](#) presents a direct comparison of the SDSS BOSS DR12 anisotropic BAO measurements [Alam et al. \[2017\]](#) with the predictions of Λ CDM and Ψ CDM. The deformation parameter ε_0 induces a smooth, monotonic, percent-level modulation across the BAO redshift range.

E Raw BAO Fits: DESI DR2

DESI DR2 [DESI Collaboration \[2025a,b\]](#) provides the most precise late-time geometric-distance measurements currently available. Figure [A4](#) shows raw DESI comparisons for D_M/r_d , D_H/r_d , and D_V/r_d . DESI’s sub-percent precision drives the mild positive preference for $\varepsilon_0 > 0$ observed in the joint likelihood (Fig. [8](#)).

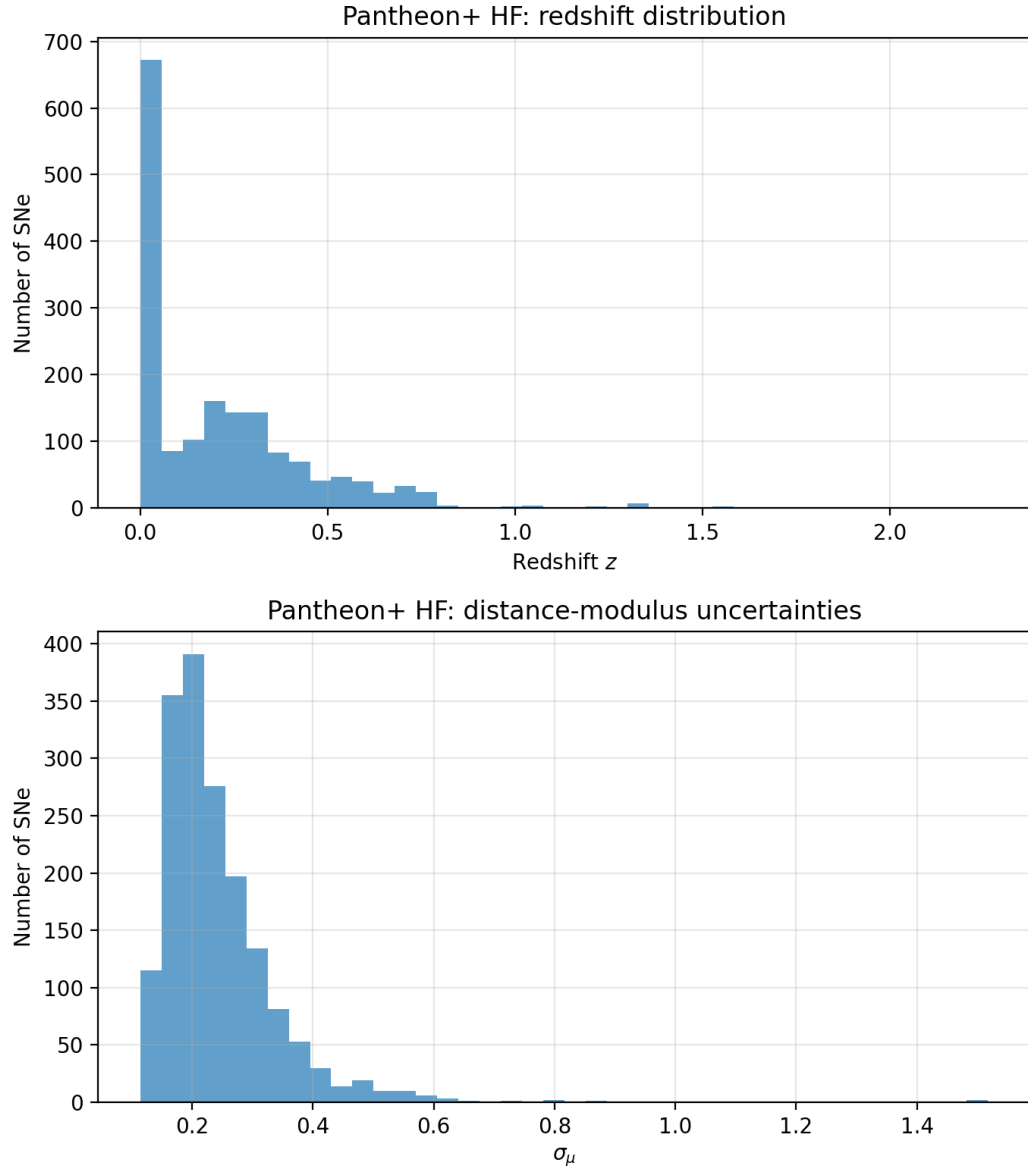


Figure A1: Diagnostic histograms for the Pantheon+ HF sample: redshift distribution, distance-modulus uncertainties, and residual structure.

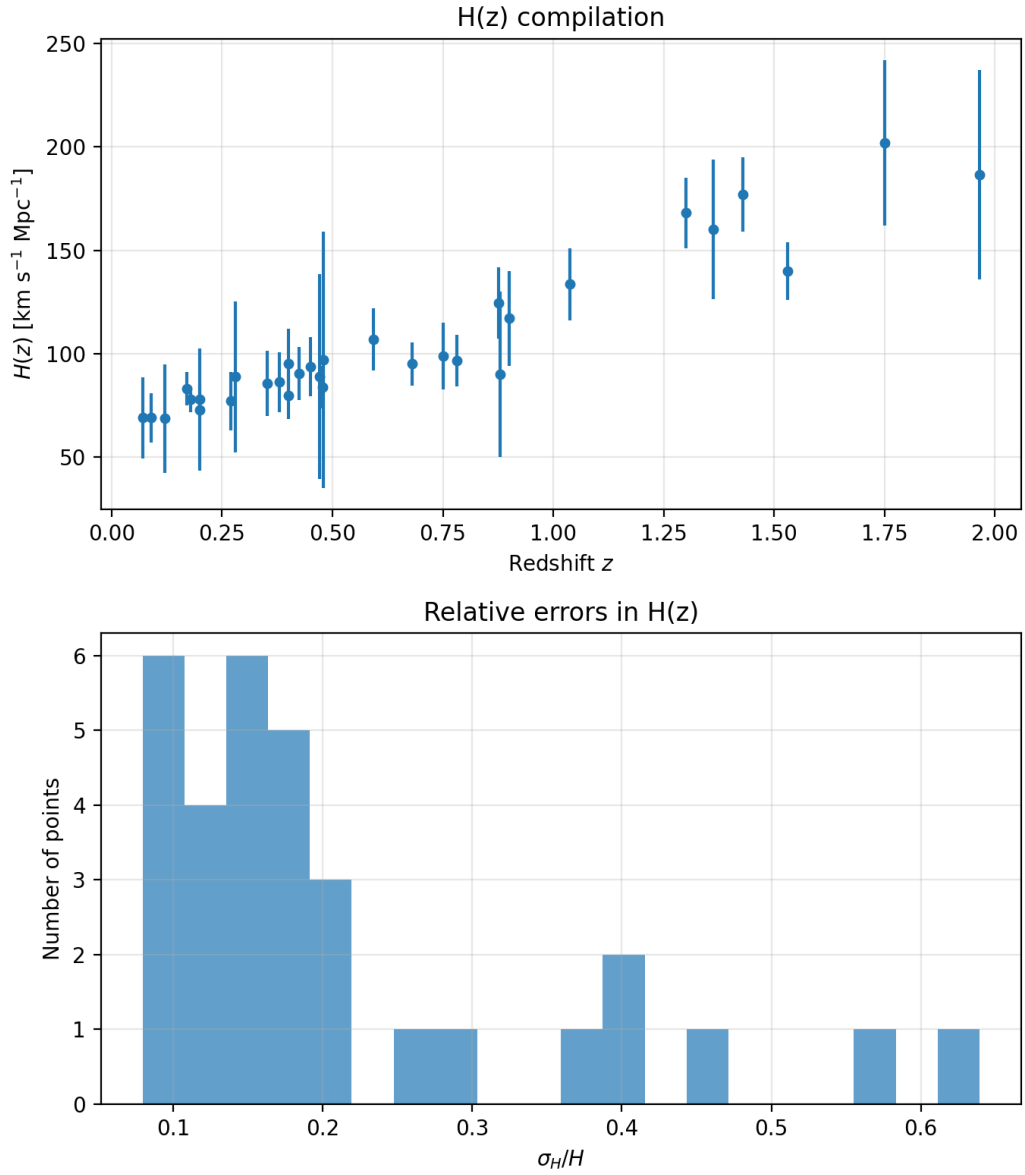


Figure A2: Relative uncertainties and redshift distribution of cosmic-chronometer data. The large measurement scatter explains their limited sensitivity to percent-level modifications of the expansion history.

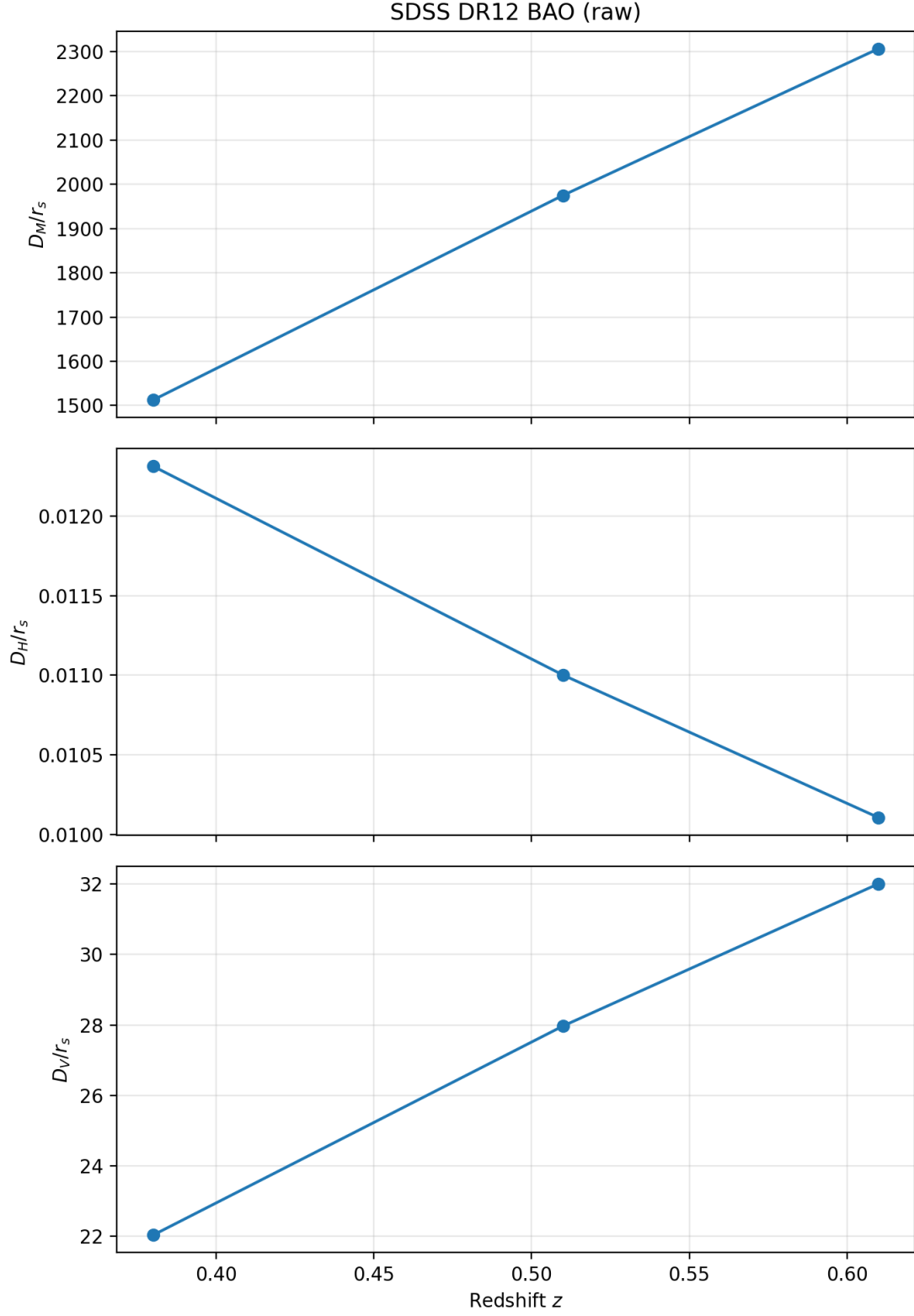


Figure A3: Raw comparison of SDSS DR12 anisotropic BAO measurements with Λ CDM and Ψ CDM predictions. The ε_0 deformation produces a smooth percent-level modulation while preserving the standard BAO structure.

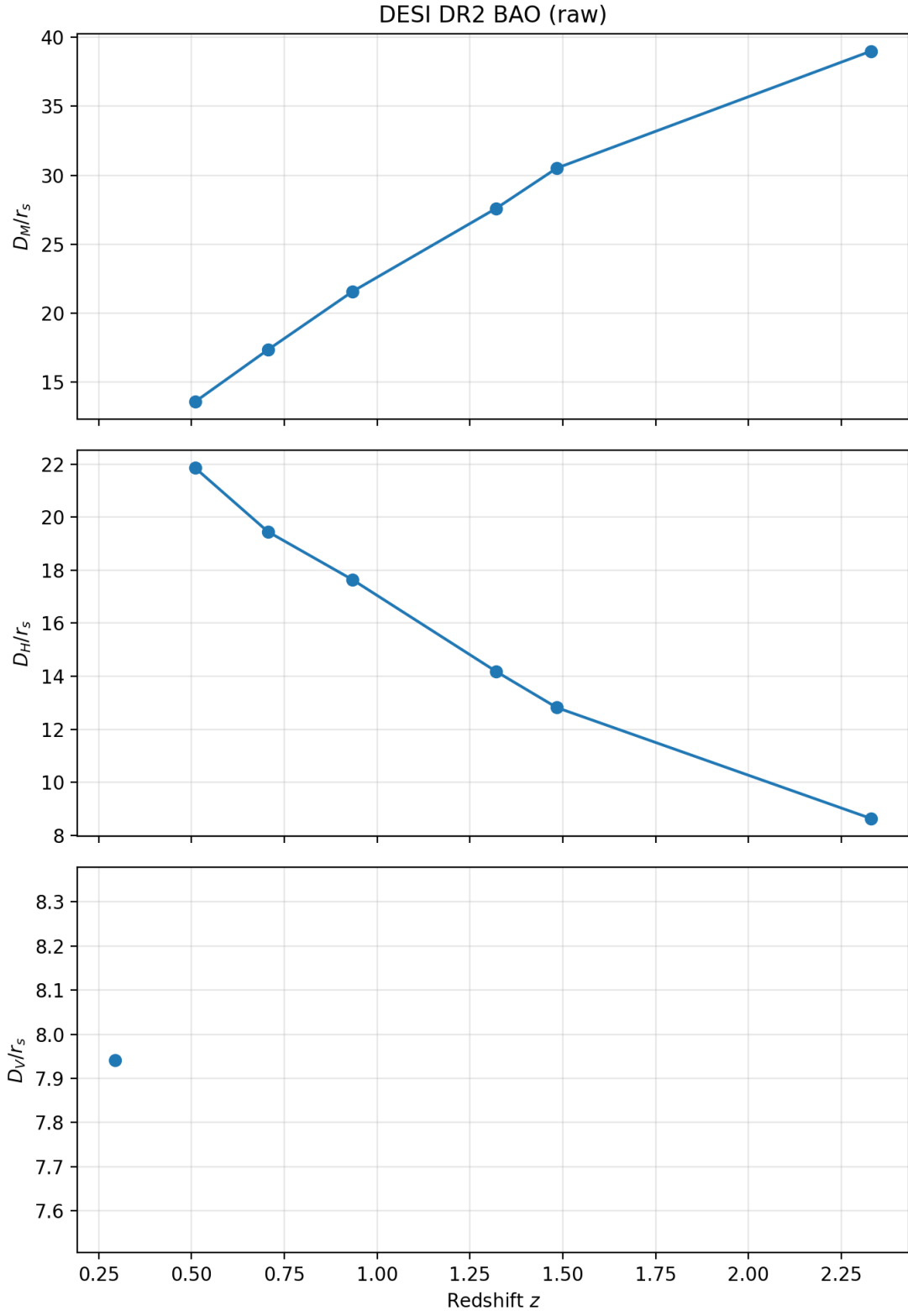


Figure A4: DESI DR2 raw BAO comparisons for D_M/r_d , D_H/r_d , and D_V/r_d . DESI's sub-percent sensitivity leads to the mild preference for $\varepsilon_0 > 0$ observed in the joint likelihood.

F Model Behaviour and Supplementary Predictions

This section complements Section 4 by presenting additional diagnostic behaviour for both Λ CDM and Ψ CDM.

G Likelihood Structure and Additional χ^2 Profiles

Table 1 summarises contributions to the total χ^2 from each dataset for both Λ CDM and Ψ CDM at $\varepsilon_0 = 0.05$. DESI DR2 provides the dominant improvement.

Table 1: Joint χ^2 contributions from each dataset at $\varepsilon_0 = 0.05$ ($H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$).

Dataset	$\chi^2_{\Lambda\text{CDM}}$	$\chi^2_{\Psi\text{CDM}}$	$\Delta\chi^2$
SN (Pantheon+ HF)	2609.197	2616.573	+7.376
$H(z)$ chronometers	11.787	11.746	−0.041
SDSS DR12 BAO	11.141	10.633	−0.508
DESI DR2 BAO	40.298	27.304	−12.995
Total	2672.423	2666.256	−6.168

Table 2: Best-fit deformation-parameter estimates from the joint likelihood and from the full ε_0 scan.

Method	Best-fit ε_0	Notes
Joint likelihood summary	0.031	Local minimum around Λ CDM
Full ε_0 scan	0.075	Global χ^2 minimum

Combined likelihood components are shown in the main text (Fig. 8). This appendix provides complementary single-dataset diagnostics for completeness.

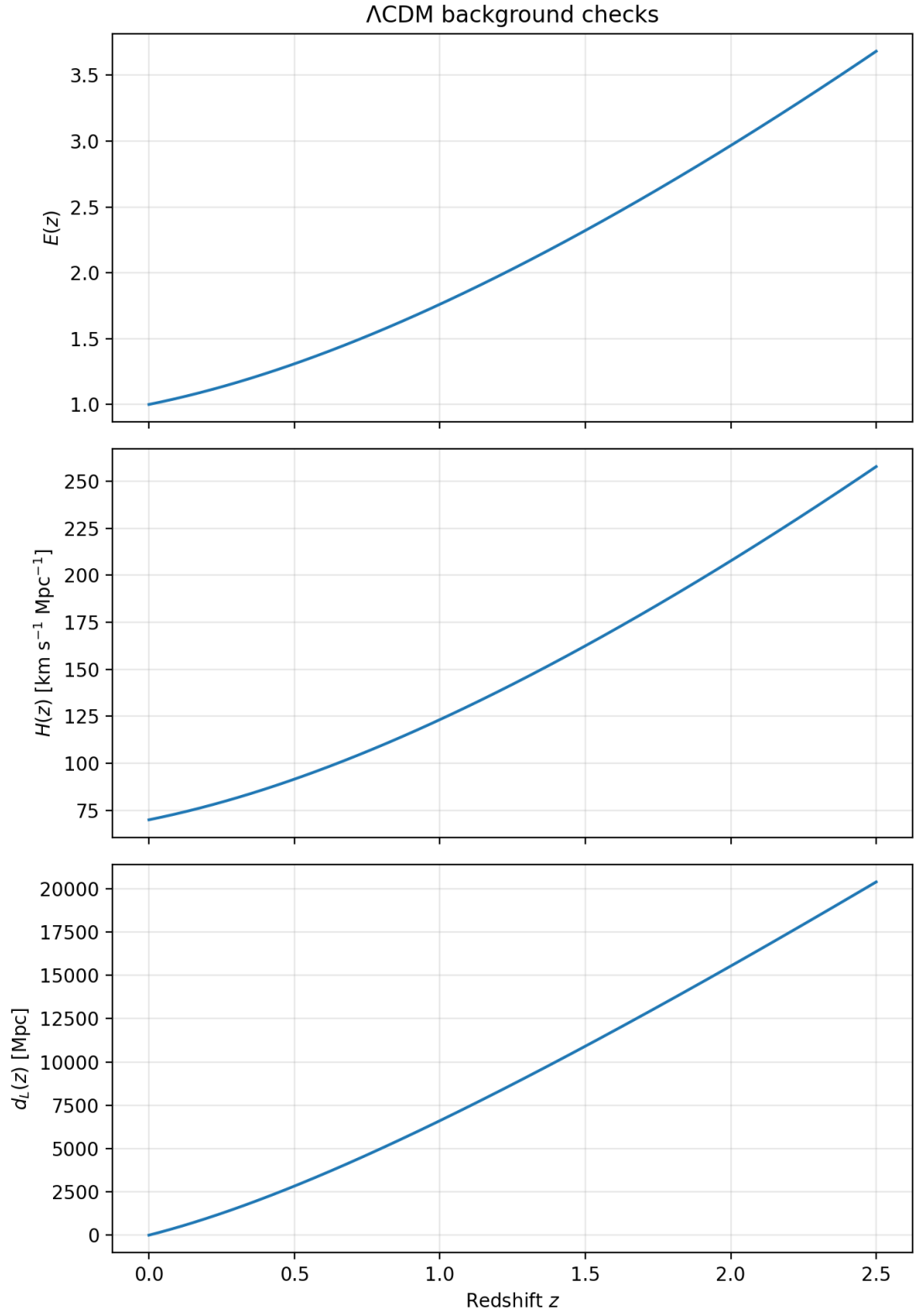


Figure A5: Baseline Λ CDM background relations: dimensionless expansion rate $E(z)$, Hubble function $H(z)$, and luminosity distance $d_L(z)$.

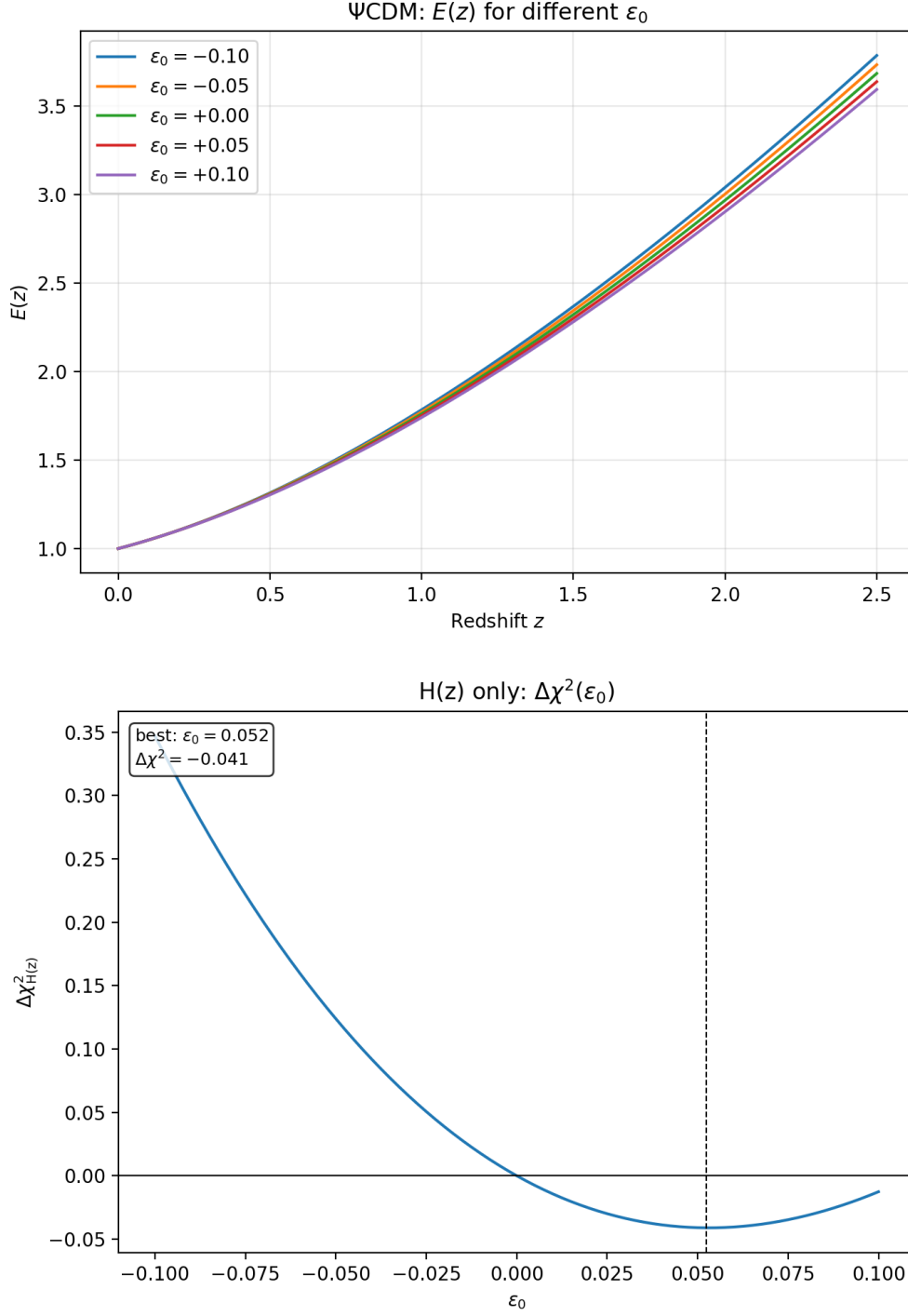


Figure A6: Supplementary model diagnostics. **Top:** dependence of the Ψ CDM expansion rate $E_\Psi(z)$ on ε_0 . **Bottom:** chronometer-only $\Delta\chi^2$ profile showing the weak constraining power of current CC data.

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